Module 8 Assignment

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. Consider the inner product space $\mathbb{F}^{m\times n}$ with inner product $\langle A, B\rangle = \text{tr}(B^*A)$. Determine the angle (in radians) between the matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

2. Let $A \in \mathbb{F}^{n\times n}$ be an invertible matrix and $\|\cdot\|$ be a norm on $\mathbb{F}^{n\times 1}$. Prove that the mapping $\|\cdot\|_A : \mathbb{F}^{n\times 1} \to \mathbb{F}$ defined by $\|x\|_A = \|Ax\|$ for $x \in \mathbb{F}^{n\times 1}$ is also a norm on $\mathbb{F}^{n\times 1}$.

3. Prove the polarization identity for real inner product spaces. That is, if V is an inner product space over $\mathbb{R}$ and $x, y \in V$, prove that $4\langle x, y\rangle = \|x + y\|^2 - \|x - y\|^2$.

4. In this problem, we will prove the maximum absolute row sum formula for $\|A\|_\infty$. Let $A \in \mathbb{F}^{m\times n}$ and let $\alpha = \max_{1\le i\le m} \sum_{j=1}^{n} |a_{ij}|$.

   (a) Show that for any $x \in \mathbb{F}^{n\times 1}$ with $\|x\|_\infty = 1$, $\|Ax\|_\infty \le \alpha$.

   (b) Suppose the kth row of A is the row with absolute sum equal to α. Find a vector $y \in \mathbb{F}^{n\times 1}$ with $\|y\|_\infty = 1$ such that $\|Ay\|_\infty = \alpha$. (**Hint.** Define the entries of y in terms of appropriate entries of $A = [a_{ij}]$. Remember that for any $z \in \mathbb{C}$, $z\overline{z} = |z|^2 = |\overline{z}|^2$ and $|z/|z|| = 1$.)

   Make sure you understand why parts (a) and (b) suffice to prove the maximum absolute row sum formula for $\|A\|_\infty$!



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- Section 4.1: 1, 2, 3, 5, 7, 8, 12
- Section 4.2: 4, 5, 6, 7, 8, 10
- Section 4.9: 7

1. Let $\|\cdot\|$ be a norm. Prove the following properties of the distance function:
   (a) $d(x, y) \geq 0$.
   (b) $d(x, y) = 0$ if and only if $x = y$.
   (c) $d(x, y) = d(y, x)$.
   (d) $d(x, z) \leq d(x, y) + d(y, z)$.
2. Compute the 1-norm, the ∞-norm, and the Frobenius norm for the following matrices.

$$\text{(a)} \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix} \qquad \text{(b)} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix} \qquad \text{(c)} \begin{bmatrix} 1 & 2 & 2 & 0 \\ 1 & -3 & 0 & -1 \\ 1 & 1 & -2 & 0 \\ -2 & 1 & 6 & 1 \end{bmatrix}$$

3. Show that $\|A\| = \max_{1 \leq i \leq m, 1 \leq j \leq n} |a_{ij}|$ is not submultiplicative. (**Hint.** Let A be a matrix whose entries are all equal to 1.)
4. Show that the Frobenius norm can be computed as $\|A\|_F = (\text{tr}(A^* A))^{1/2}$ or $\|A\|_F = (\text{tr}(AA^*))^{1/2}$
5. Let A, B be matrices and x be a vector such that AB and Ax are defined.
   (a) Show that $\|Ax\|_2^2 \leq \|A\|_F \, \|x\|_2$.
   (b) Show that $\|AB\|_F^2 \leq \|A\|_F^2 \, \|B\|_F^2$. This shows that the Frobenius norm is submultiplicative and thus a matrix norm.
6. (a) Show that $\|I\| = 1$ for every induced matrix norm.
   (b) What is $\|I_{n \times n}\|_F$? (This shows the Frobenius norm is not an induced norm!)

